

Problem set 4 / Linear transformations

A short course in the relationships behind the results.

Problem 4.1. A linear map in local notation

Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear. Prove that $\dim \ker T + \dim \operatorname{im} T = n$.

Show that the claim follows from the dimension theorem.

Definition 3.2. Linear maps

A map T is linear when $T(au + bv) = aT(u) + bT(v)$.

The argument follows from Theorem 3.5.

Definition 3.3. Kernel and image

The kernel is the set of vectors sent to zero. The image is the range of T .

The argument follows from Theorem 3.6.

Theorem 3.4. Rank-Nullity

(i) The kernel and image of A are subspaces. (ii) If V is finite dimensional, $\dim \ker A + \dim \operatorname{im} A = \dim V$.

The argument follows from Theorem 3.7.

Problem set 4 / A proof in context

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Corollary 3.5. Injectivity criterion

A linear map is injective if and only if its kernel contains only zero.

The argument follows from Theorem 3.8.

Lemma 3.6. Basis extension

Every linearly independent list extends to a basis of a finite dimensional space.

The argument follows from Theorem 3.9.

Definition 3.7. Direct sums

If U and W intersect only at zero, their sum is a direct sum.

The argument follows from Theorem 3.10.

Definition 3.8. Isomorphisms

An invertible linear map identifies two vector spaces of equal dimension.

The argument follows from Theorem 3.11.

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Proposition 3.9. Composition

The composition of two linear maps is linear.

The argument follows from Theorem 3.12.

Theorem 3.10. Dimension formula

The dimension of $U + W$ equals $\dim U + \dim W$ minus $\dim$ of their intersection.

The argument follows from Theorem 3.1.

Definition 3.11. Quotient spaces

The quotient V/U consists of cosets of U , with the induced vector operations.

The argument follows from Theorem 3.2.

Theorem 3.12. First isomorphism theorem

The quotient of V by $\ker T$ is isomorphic to the image of T .

The argument follows from Theorem 3.3.